Northern
Colorado
Geotech

2956 29th Street, Unit 21
Greeley, Colorado 80631
Phone: (970) 506-9244
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August 12, 2022

Rockton 275 Holbrook, LLC
2801 North 111th Street
Erie, Colorado 80516

Attn: Mark Rockwell

Re: Subsurface Exploration
275 Holbrook Street
Erie, Colorado
Northern Colorado Geotech Project No. 115-22

On July 7, 2022 Northern Colorado Geotech performed a subsurface exploration at the above referenced address in Erie, Colorado. The purpose of the exploration was to provide a site specific geotechnical exploration for the proposed garage to be built on the lot.

One test bore was drilled near the center of the proposed garage to a depth of 15 feet. The soils encountered at the site consisted of silty sand, sandy lean clay and well graded sand with silt and gravel to a depth of 12 feet. The soils were underlain by siltstone/claystone bedrock. Groundwater was encountered at a depth of 9 feet.

Foundation Design

Based on the anticipated construction and the results of our exploration, it is our opinion that the proposed garage may be supported on a spread footing foundation system. The footings may be designed for a maximum bearing pressure of 1,500 psf. The design bearing pressure applies to dead loads plus design live load conditions. The design bearing pressure may be increased by one-third when considering total loads that include wind or seismic conditions.

Exterior footings should be placed a minimum of 30 inches below finished grade for frost protection and to provide confinement for the bearing soils. Finished grade is the lowest adjacent grade for perimeter footings.

Foundations and masonry walls should be reinforced as necessary to reduce the potential for distress caused by differential foundation movement. The use of joints at openings or other discontinuities in masonry walls is recommended.

Foundation excavations should be observed by Northern Colorado Geotech. If the soil conditions encountered differ significantly from those presented in this report, supplemental recommendations may be required.

Floor Slab Design and Construction

Some differential movement of slab-on-grade floor systems is possible should the subgrade soils become elevated in moisture content.

Additional floor slab design and construction recommendations are as follows:

- Positive separations and/or isolation joints should be provided between slabs and all foundations, columns or utility lines to allow independent movement.
- Control joints should be provided in slabs to control the location and extent of cracking.
- Interior trench backfill placed beneath slabs should be compacted to 95 percent of ASTM D698.
- In areas subjected to normal loading, a minimum 4-inch layer of sand, clean-graded gravel or aggregate base course should be placed beneath interior slabs.
- Floor slabs should not be constructed on frozen subgrade.

General Earthwork

All earthwork on the project should be observed and evaluated by Northern Colorado Geotech. The evaluation of earthwork should include observation and testing of engineered fill, subgrade preparation, foundation bearing soils, and other geotechnical conditions exposed during the construction of the project.

Site Preparation

Strip and remove existing vegetation, debris, and other deleterious materials from proposed building and pavement areas. All exposed surfaces should be free of mounds and depressions which could prevent uniform compaction.

Stripped materials consisting of vegetation and organic materials should be wasted from the site, or used to revegetate landscaped areas or exposed slopes after completion of grading operations.

If unexpected fills or underground facilities are encountered, such features should be removed and the excavation thoroughly cleaned prior to backfill placement and/or construction.

It is anticipated that excavations for the proposed construction can be accomplished with conventional earthmoving equipment.

The individual contractor(s) is responsible for designing and constructing stable, temporary excavations as required to maintain stability of both the excavation sides and bottom. All excavations should be sloped or shored in the interest of safety following local, and federal regulations, including current OSHA excavation and trench safety standards.

Fill Materials and Placement

All exposed areas which will receive fill should be scarified to a minimum depth of eight inches, conditioned to near optimum moisture content, and compacted.

The placement of soils on the site should be observed by Northern Colorado Geotech. The fill should be assessed for suitability of use in the proposed fill and tested for placement including compaction percentage and moisture content.

Engineered fill should be placed and compacted in horizontal lifts, using equipment and procedures that will produce recommended moisture contents and densities throughout the lift. Recommended compaction criteria for engineered fill materials are as follows:

Clean on-site soils or approved imported materials may be used as fill material.

Imported soils (if required) should conform to the following:

Gradation	**Percent fines by weight (ASTM C136)**
6"	100
3"	70-100
No. 4 Sieve	50-100
No. 200 Sieve	50 (max)
Liquid Limit	30 (max)
Plasticity Index	5 (max)

Material	**Minimum Percent (ASTM D698)**
Scarified subgrade soils	95
On-site and imported fill soils:	
Beneath foundations	95
Beneath slabs	95

On-site or imported granular soils should be compacted within a moisture range of 3 percent below to 3 percent above optimum unless modified by the project geotechnical engineer.

GENERAL COMMENTS

The analysis and recommendations presented in this report are based upon data obtained from one boring performed to obtain representative subsurface conditions at the site. Variations in the soil at the site will occur. Northern Colorado Geotech should be present during construction to observe the excavation and construction procedures and confirm or modify our recommendations.

The scope of services for this project does not include either specifically or by implication any environmental assessment of the site.

This report is intended exclusively for the use by the client. Any use or reuse of the findings and/or recommendations of this report by parties other than the client without the written consent of Northern Colorado Geotech is undertaken at said parties' sole risk.

Subsurface Exploration
275 Holbrook – Erie, Colorado
Northern Colorado Geotech Project No. 115-22

This report has been prepared in accordance with generally accepted geotechnical engineering practices in this area at this time. No warranties, either express or implied, are intended or made.

If you have any questions concerning this report or any of our consulting services, please do not hesitate to contact us at (970) 506-9244.

Sincerely,
NORTHERN COLORADO GEOTECH

Prepared by:
Doug Leafgren
Digitally signed by Doug Leafgren
DN: C=US, OU=President, O=Northern Colorado Geotech, CN=Doug Leafgren, E=dleafgren@msn.com
Reason: I am the author of this document
Location: your signing location here
Date: 2022.08.12 11:10:02-06'00'
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Gary G. Weeks, P.E.
Vice President

Copies to: Addressee (2)

Attachments: Log of Boring
Laboratory Test Results

LOG OF BORING No. 1

CLIENT **Rockton 275 Holbrook**	ARCHITECT/ENGINEER
SITE **275 Holbrook Street Erie, Colorado**	PROJECT **Proposed Garage with Living Quarters**

		SAMPLES				TESTS			
	GRAPHIC LOG / DEPTH (FT.)	BLOWS/12" N-VALUE	NUMBER	TYPE	IN. DRIVEN IN. RECOVERED	MOISTURE, %	DRY DENSITY PCF	HAND PENE-TROMETER psf	LIQUID LIMIT PLASTIC INDEX PERCENT FINES
0.5 6" TOPSOIL SILTY SAND Brown, moist, loose to medium dense		14	1	SS	18	4			
		14	2	RS	12	6	106	3,000	0% Swell
5.0 SANDY LEAN CLAY Tan, moist, medium dense	5								
9.0 WELL GRADED SAND WITH SILT AND GRAVEL Brown, moist to wet, medium dense	10	42	3	RS	12	8			
12.0 WEATHERED SILTSTONE/CLAYSTONE Dark, brown, moist, soft		19	4	SS	12	21			
15.0 BOTTOM OF BORING	15								

WATER LEVEL OBSERVATIONS			
WL	▽	9.0	W.D.
WL	▼	9.0	A.B.
When Checked 1 Day A.B.			

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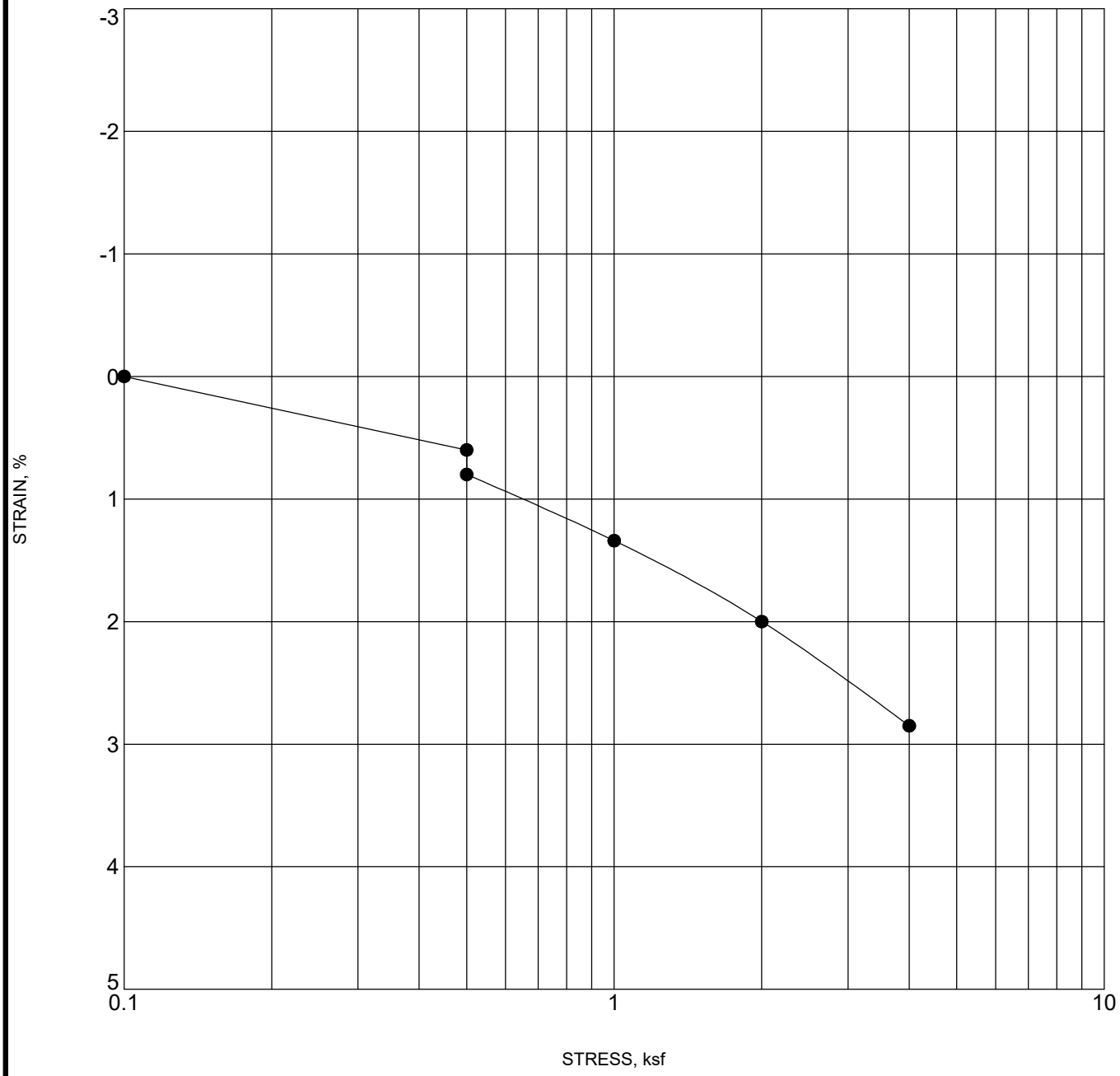
STARTED	7/7/22	FINISHED	7/7/22
DRILL CO.	Drilling Eng	DRILL RIG	CME-55
LOGGED BY	TK	APPROVED	DML
NCG PROJECT NO.	115-22		



Specimen Identification			Classification	γ_d	MC%
●	1	4.0	**Silty Sand**	**106**	**6**

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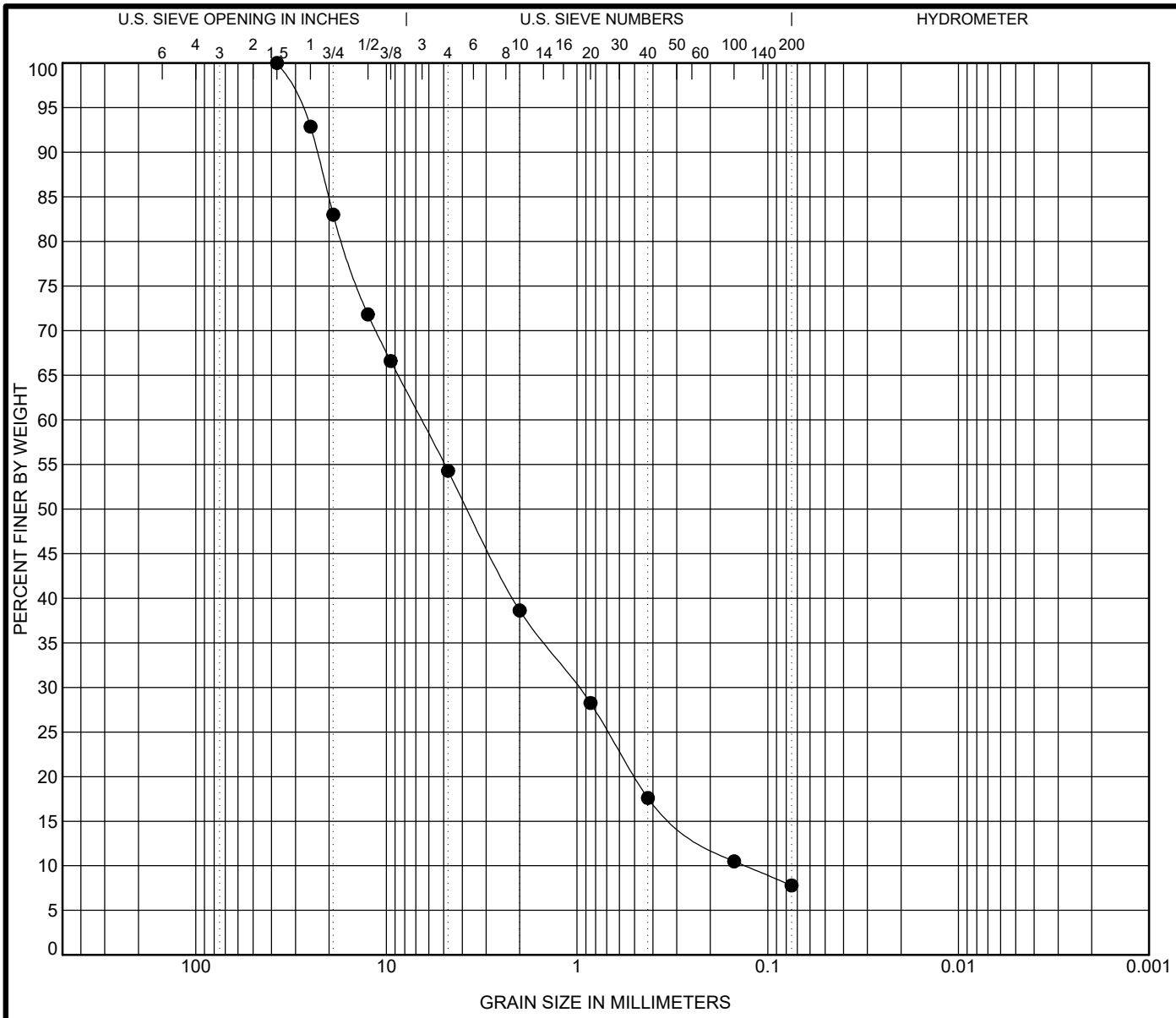
CONSOLIDATION TEST

Client: Rockton 275 Holbrook Number: 115-22

Project: Proposed Garage with Living Quarters

Location: 275 Holbrook Street

115-22.GPJ



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Specimen Identification		Classification	LL	PL	PI	Cc	Cu
●	**1**	**9.0**	**Well Graded Sand with Silt and Gravel**				**1.11**	**49.64**

	Specimen Identification		D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	**1**	**9.0**	**37.5**	**6.556**	**0.982**	**0.132**	**45.7**	**46.5**	**7.8**	

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GRAIN SIZE DISTRIBUTION

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115-22.GPJ